

SCB ModelGuard—— Cross-Border Credit Model Compliance Radar

Presented by Chen Shiyu

Why this product is right for our company

- **Meridian Model Assurance** — growth-stage RegTech (~35 people, Series A)
- **Mission:** Make cross-border model risk visible before regulators and customers pay the price
- **What we do best:** Jurisdiction rule engines + quantitative drift monitoring (not policy PDFs)
- **Why now:** Standard Chartered runs SCB-CS-001 in the US and Singapore under diverging 2026 rules
- **Commercial fit:** One platform → two defensible committee packs → one anchor-bank contract (target USD 400k–1.2M ACV)



The problem spreadsheets and memos cannot solve

- Static Excel / Word memos show last quarter's approval rate
- They cannot answer: "The economy changed—does the same score still mean the same risk?"
- 2022–23 → 2025–26 (demo portfolio):
 - Median income ~-11%
 - Unemployment ~2.8% → 6.7%
 - Portfolio default ~25% → 35%
 - Approvals ~99% → 90%
- Spreadsheets show that approvals fell—not why the scoring engine may mislead the risk committee



What SCB ModelGuard does

- SCB ModelGuard monitors whether SCB-CS-001 still ranks risk correctly, whether the applicant pool has shifted, and—where law allows—whether outcomes remain fair.
- Then translates the same metrics into US (OCC) vs Singapore (MAS) compliance language: **PASS / WARN / FAIL**

01. CSV data
(train + live)



02. Metrics engine
(AUC, PSI, ARR)



03. Dual reports
(US_OCC / SG_MAS)

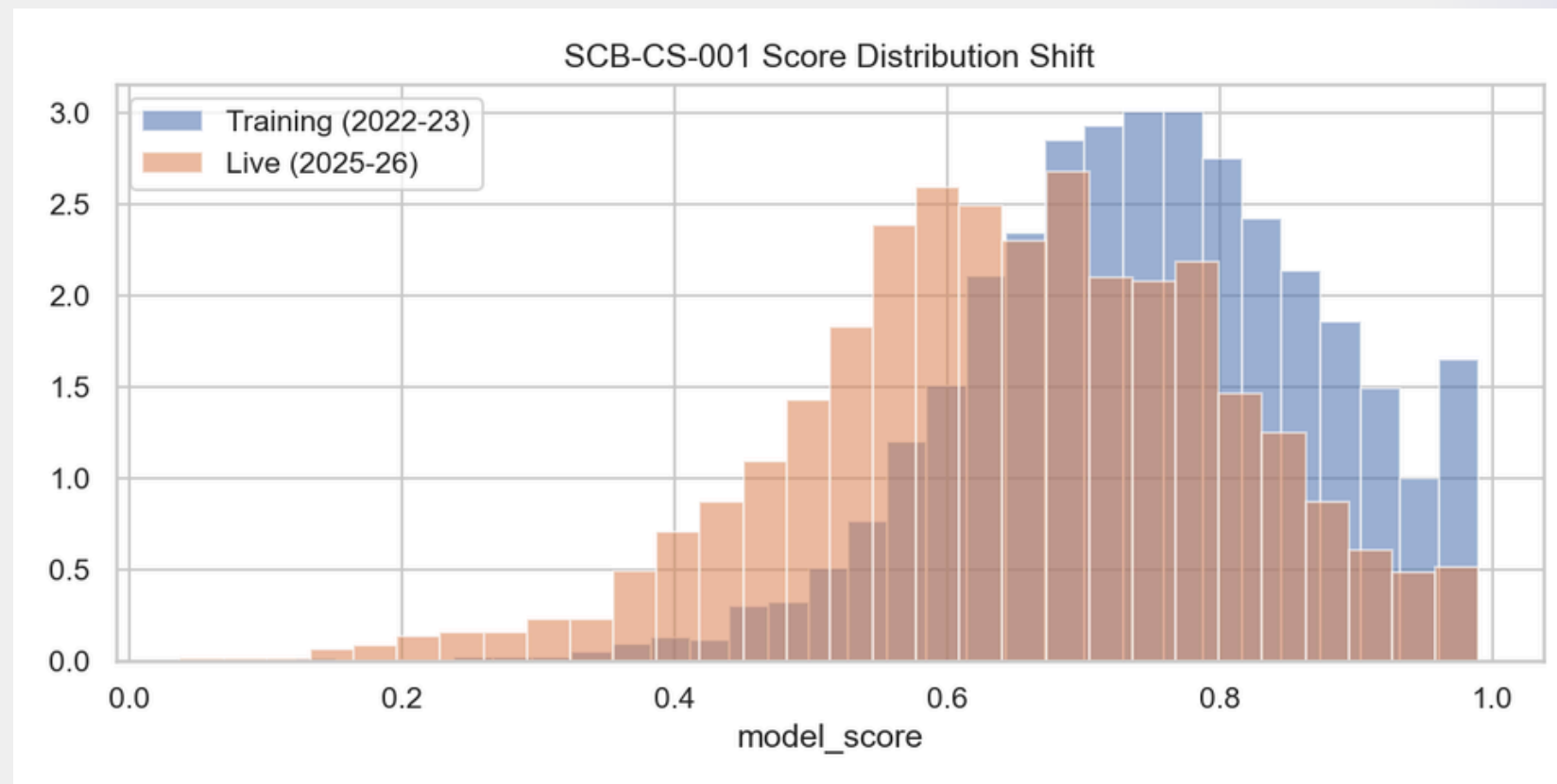


Regulatory divergence: what we handle and how

	United States (OCC 2026-13)	Singapore (MAS / FEAT)
Model	Traditional PD score in scope (GenAI excluded, not this model)	Same model
Fairness data	Ethnicity stripped at ingest (ECOA / fairness-under-unawareness)	Native ethnicity used for outcome monitoring
Drift (PSI ~0.46)	WARN (FAIL bar higher: principles-based)	FAIL (stricter threshold)
Typical action	Independent model validation review	Pause rollout expansion + committee pack



Live results on the same book of business



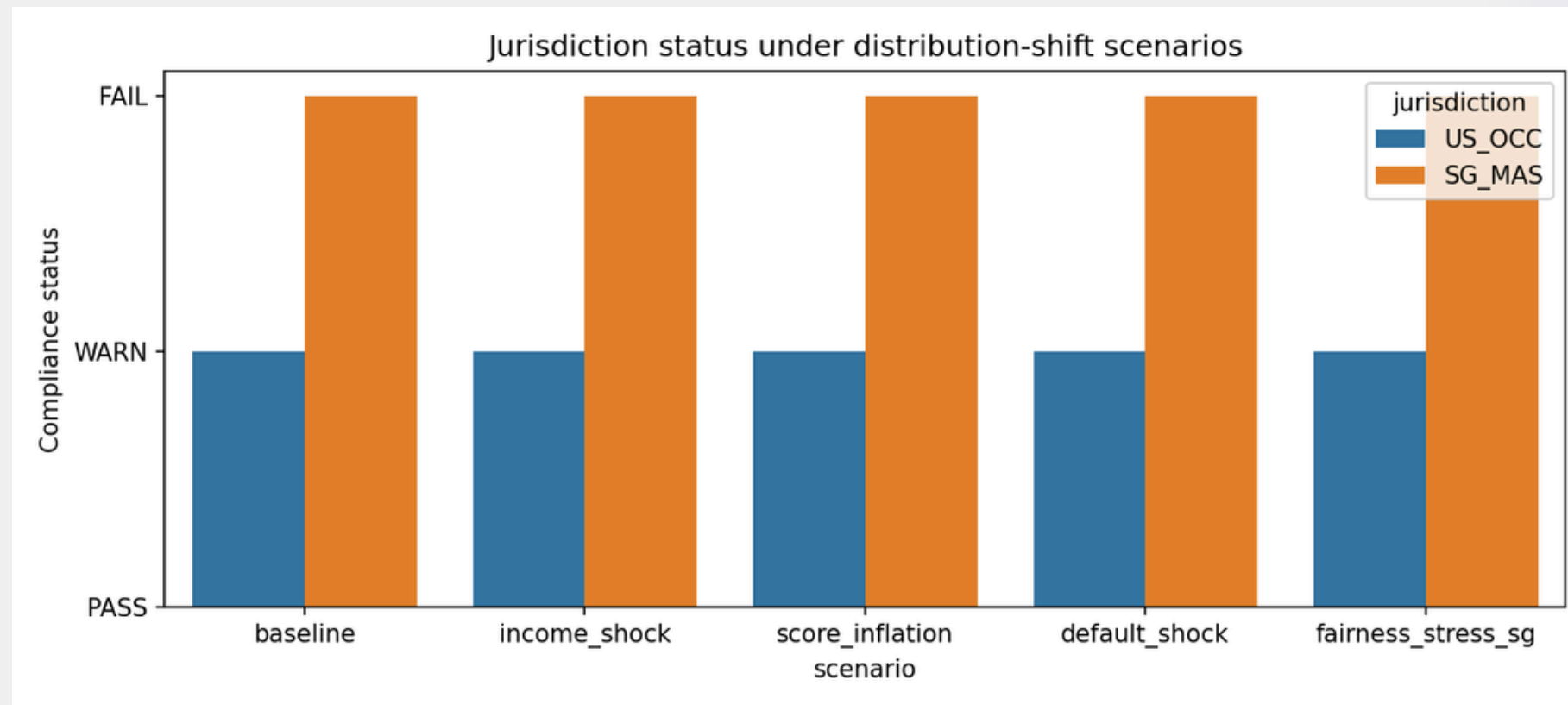
United States — ● **WARN**

- $\Delta\text{AUC} \approx 0.015$ (rank-order still modestly stable)
- $\text{PSI} \approx 0.46$ (material population shift)
- Fairness: advisory only (no auto-fail on ethnicity)
- Action: Escalate independent validation; monitor state litigation / ESG separately

Singapore — ● **FAIL**

- $\text{PSI} > 0.25$; low-score share +4.8 pp; default rate +9.8 pp
- Action: Stop expanding live volume; root-cause + FEAT outcome review

Stress tests: the flip is structural, not accidental



- Scenarios tested: income shock, score inflation, default shock, fairness stress (SG path)
- Every scenario: US stays WARN, Singapore stays FAIL
- Conclusion: Jurisdiction logic is built into rule design, not a one-off tuning exercise

06

Judgement cannot be outsourced

humans must decide:

1. **Cut-off** adjustment on model scores
2. **Full model retrain** (USD millions, 6–12 months)
3. **Proactive MAS notification** or not
4. **US BISG exploratory** fair-lending audit—and whether it goes to the board
5. **Singapore FAIL:** halt growth vs enhanced monitoring only
6. Never re-insert **ethnicity into US production** just to “green” a global dashboard

07

What we deliberately do not do — and why

We do not	Why (boundary choice)
Auto-approve or auto-decline loans	Credit judgement and risk appetite are human
Issue US “fair lending cleared” badges from ethnicity	Post-Reg B trajectory + ECOA unawareness—would be documentation theatre
Replace lawyers on state fair-lending exposure	Tool is risk radar, not legal counsel
Auto-email MAS	No regulatory API; governance is a human choice
Claim synthetic demo data = production SCB book	Honest scope limit for this class submission

If we had more time / real data: BISG exploratory module, adverse-action explainability, versioned rule packs, live feed integration.



Thank You